

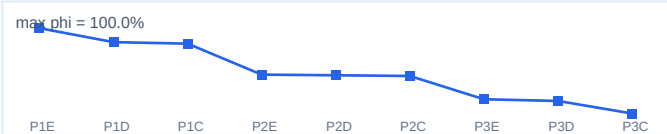
Khosla Hydraulic Foundation Designer

One-page hydraulic foundation seepage and uplift pressure design summary

UNSAFE

PROJECT Khosla Design Example	STRUCTURE Barrage / Weir Floor	DESIGNER -	DATE 2026-07-06
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SEEPAGE HEAD H 5.000 m	MAX UPLIFT PRESSURE 49.05 kN/m2	CRITICAL POINT Pile 1 E	EXIT GRADIENT GE 0.12900	REQUIRED MAX T 3.571 m
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HYDRAULIC AND GEOMETRY INPUTS	
Pond level	105.000 m
Tail water level	100.000 m
Safe exit gradient	0.16670
Floor length b	72.000 m
Crest RL	102.000 m
Floor RL	99.000 m
CORRECTED PRESSURE PROFILE	
	

FLOOR THICKNESS NODES		
Node	x (m)	t (m)
N1	0.00	3.00
N2	42.00	4.00
N3	72.00	2.00

PILE SCHEDULE			
Pile	x	d	t
P1	0.00	5.00	3.00
P2	42.00	6.00	4.00
P3	72.00	4.00	2.00

SAFETY SUMMARY AND RECOMMENDATION	
Overall design status: UNSAFE Exit-gradient status: SAFE Unsafe pressure points: 1 Recommendation: Review unsafe points and increase floor thickness or pile depth where required	

UPLIFT PRESSURE AND FLOOR SAFETY RESULTS													
Pile	Pt	x	d	Base	Mut.	t corr.	Slope	Final	h	P	Req t	Prov t	Status
P1	E	0.0	5.0	100.0	0.0	0.0	0.0	100.0	5.00	49.1	3.57	3.00	UNSAFE
P1	D	0.0	5.0	83.6	0.0	0.0	0.0	83.6	4.18	41.0	2.99	3.00	SAFE
P1	C	0.0	5.0	76.6	1.1	4.2	0.0	81.9	4.09	40.2	2.92	3.00	SAFE
P2	E	42.0	6.0	50.1	-1.0	-3.5	0.0	45.6	2.28	22.4	1.63	4.00	SAFE
P2	D	42.0	6.0	44.8	0.0	0.0	0.0	44.8	2.24	22.0	1.60	4.00	SAFE
P2	C	42.0	6.0	39.4	1.0	3.6	0.0	44.0	2.20	21.6	1.57	4.00	SAFE
P3	E	72.0	4.0	21.0	-1.2	-3.1	0.0	16.7	0.83	8.2	0.60	2.00	SAFE
P3	D	72.0	4.0	14.7	0.0	0.0	0.0	14.7	0.74	7.2	0.53	2.00	SAFE
P3	C	72.0	4.0	0.0	0.0	0.0	0.0	0.0	0.00	0.0	0.00	2.00	SAFE